Enrolment No./Seat No_____________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–VII (NEW) EXAMINATION – SUMMER 2024

Subject Code:3170723 Date:15-05-2024

Subject Name:Natural Language Processing

Time:02:30 PM TO 05:00 PM Total Marks:70

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			MARKS
Q.1	**(a)**	Briefly describe the applications of NLP in real life.	**03**
	(b)	Write a note on how to build NLP Pipeline.	**04**
	(c)	Provide a detailed explanation of the various phases in Natural Language Processing.	**07**
Q.2	**(a)**	Describe Kneser-Ney smoothing with suitable example.	**03**
	(b)	Draw and explain architecture of NLG system.	**04**
	(c)	Explain Unigram, Bi-gram and Tri-gram concepts using an example sentence: “The purpose of our life is to happy”.	**07**
		OR	
	(c)	Describe Part-of-Speech Tagging with suitable example.	**07**
Q.3	**(a)**	Explain word embedding in brief.	**03**
	(b)	Describe Embedding representations for words Lexical Semantics with example.	**04**
	(c)	Explain Word Sense Disambiguation with Lesk algorithm.	**07**
		OR	
Q.3	**(a)**	Draw the dependency tree for following sentence: “Jack ate an apple yesterday which was red”	**03**
	(b)	Explain “Continuous bag of words” with an example.	**04**
	(c)	Write a note on comparison of various smoothing Techniques.	**07**
Q.4	**(a)**	Write a short note on Named Entity Recognition.	**03**
	(b)	State different algorithms used for relation extraction.	**04**
	(c)	Explain stages of IR-based question answering models.	**07**
		OR	
Q.4	**(a)**	Explain sentiment mining in brief.	**03**
	(b)	How is Text Classification and Text Summarization works in NLP?	**04**
	(c)	Explain the concept of Cross-Lingual Information Retrieval in detail.	**07**
Q.5	**(a)**	Describe Knowledge Based Machine Translation System.	**03**
	(b)	What is Machine Translation? What are the applications of Machine Translation?	**04**
	(c)	Illustrate and provide a comprehensive explanation of the Encoder-Decoder architecture.	**07**

OR

Q.5	**(a)**	Provide a concise explanation of translation divergence.	**03**
	(b)	Describe Rule Based Machine Translation System.	**04**
	(c)	Elaborate on how parameter learning is conducted in Statistical Machine Translation (SMT) with an example.	**07**

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